MOHAMMAD ZAID MOONSAMY

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GitHub LinkedIn Google Scholar

RESEARCH INTERESTS

My research interests include artificial intelligence, scientific discovery, and global health, with a particular interest in applying machine learning and large language models to real-world healthcare challenges. I am fascinated by interdisciplinary research and the creation of new knowledge. I enjoy solving challenging problems and applying critical thinking to complex questions.

EDUCATION

Master of Science in Computer Science (by Dissertation) **Jan 2025 - Present**
University of the Witwatersand

- Thesis on “Applying AI for scientific discovery of novel Microbe-Cancer Pairs for cancer prevention.” supervised by Prof Bruce Bassett, Prof Richard Klein and Dr Hairong Wang
- Paper to be published at SAICSIT 2026 conference

Bachelor of Science Honours in Computer Science (Cum Laude) **Jan 2024 - Nov 2024**
University of the Witwatersrand

- Coursework (courses include: NLP, Computer Vision and Probabilistic Graphical Models)
- Research Project: Semi-automated grading of free-text computer science answers

Bachelor of Science in Computer Science (A aggregate) **Jan 2021 - Nov 2023**
University of the Witwatersrand

- Coursework (courses include: Mathematics, Computer Science and Applied Mathematics)

WORK EXPERIENCE

Summer@EPFL (École Polytechnique Fédérale de Lausanne) intern, Switzerland **June 2025 - Aug 2025**

- Worked under the supervision of Dr Trevor Brokowski at the Laboratory for Intelligent and Humanitarian Global Technologies (LiGHT Lab).
- Contributed to the Mam-AI project, aimed at reducing maternal mortality in Zanzibar, by developing LLM- and RAG-based solutions for edge-device deployment and a mobile application to support midwives during antenatal and postnatal care.

Consultant, GRAI Labs **Apr 2025 - May 2025**

- Worked on a Medical AI project focused on prompt optimization for Vision-Language Models in tuberculosis classification.

Junior Software Developer **June 2023 - Mar 2024**

- Worked as a junior developer to build a full-stack application for EdX using FlutterFlow (low-code platform) for the frontend and Firebase for backend services.

Computer Science Tutor, University of the Witwatersrand **Jan 2023 - Present**

- Tutored courses in NLP, Data Structures, Algorithms, Databases, Machine Learning and Artificial Intelligence, with responsibilities including tutorials, marking, and exam invigilation.

VOLUNTEER WORK

- Assisted the school admission staff with grade R and grade 1 online admissions as a part of the transition to the online remote services due to COVID-19 (60 hours, 2021-2023)
- Facilitator for “Building a Transformer from Scratch” online workshop hosted by Wits MIND institute and Kempner Institute at Harvard University (2025)

ACHIEVEMENTS

- MSc Fellowship from IDORI (2025 - 2026)
- Postgraduate Merit Award (2023 - 2026)
- Dean's list (2021 and 2024)
- University Council Merit Scholarship (2022)
- Golden Key Honour Society Member (2021)
- University Entrance Scholarship (2021)
- Finalist Wits Mathematics Comptetion (2021)

HOBBIES AND INTERESTS

- Hackathons: Have participated in various hackathons including Wits HPC (High Performance Computing) hackathon, Adapt IT hackathon, Robocup hackathon and more.
- Travelling: I enjoy exploring new places, cultures, and experiences.
- Hiking, running, and general fitness: I enjoy staying active through outdoor and physical activities.
- Photography: I enjoy capturing well-composed and meaningful shots.
- Sketching and drawing: I enjoy creative expression through visual art.
- Sports and gaming: Enthusiastic about both video games and board games.